

Real-Time Facial Animation Generation on Face Mask

Bin Han
Korea Institute of Science and
Technology
Seoul, Republic of Korea
hanbin22@kist.re.kr

Gerard Jounghyun Kim
Korea University
Seoul, Republic of Korea
gjkim@korea.ac.kr

Jae-In Hwang
Korea Institute of Science and
Technology
Seoul, Republic of Korea
hji@kist.re.kr

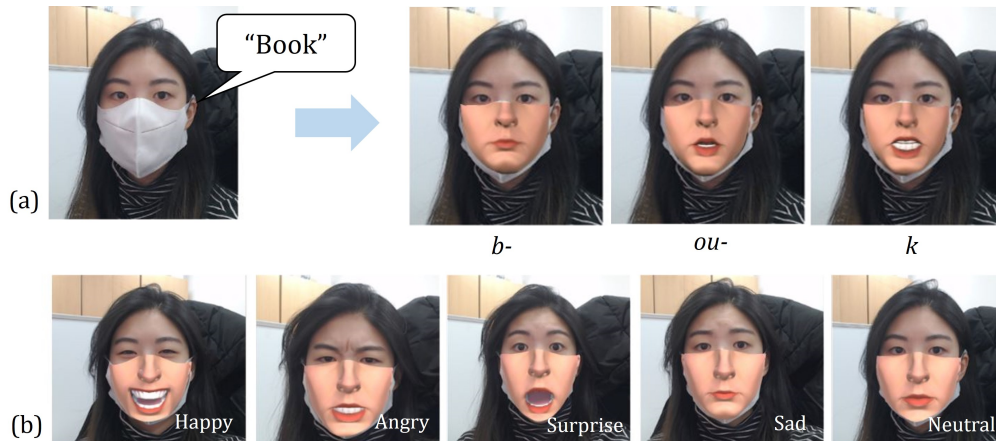


Figure 1: (a) Lip-sync animation (b) five emotional lower face animations (happy, angry, surprised, sad, and neutral)

ABSTRACT

In light of the COVID-19 pandemic, wearing a mask is crucial to avoid contracting infectious diseases. However, wearing a mask is known to impair communication functions. This study aims to address the communication difficulties caused by wearing a mask and provide a strategy for aiding in understanding the speaker's speech through facial animation. Facial animation is generated in real-time, and upper facial information is processed to detect the speaker's emotions, generating a lower facial expression. In addition, the system detects the mask's shape and enables accurate registration in the proper position. This technology can improve communication and alleviate challenges associated with communication between persons wearing face masks.

CCS CONCEPTS

• **Human-centered computing** → *Accessibility technologies*; Collaborative and social computing.

KEYWORDS

lip-sync animation, facial animation, face detection

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1 INTRODUCTION

Wearing masks has become standard practice worldwide in response to the COVID-19 pandemic. Wearing a mask creates a challenge in that communication becomes more difficult because the speaker's mouth and expression cannot be seen accurately. The issues that arise from wearing a mask might be classified as verbal or non-verbal communication issues. Non-verbal communication encompasses all forms of human communication that do not include direct linguistic expression. Examples include gaze processing, hand gestures, and body language. Usually, there is a means to convey one's thoughts through facial expressions. The form of the mouth plays a crucial role in the transmission of facial expressions and emotions [Mehrabian 2017]. Wearing a mask obstructs the transmission of sufficient information about the other person's face and makes it harder to understand their emotions [Wegrzyn et al. 2017]. Owing to this interference in information delivery, facial expressions and emotions cannot be easily conveyed, and communication issues reduction might be amplified [Carbon 2020].

This study focuses on communicating successfully by synthesizing the lower face animation. The system was developed to create 1) the appropriate facial expressions representing users' emotions by analyzing upper face information, 2) generate lip-sync animations by analyzing sound, and after 3) combine them at proper positions using face detection information.

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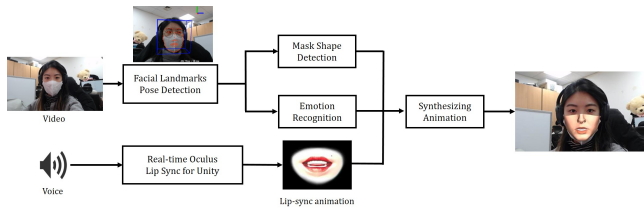


Figure 2: Overall system design

2 RELATED WORKS

In recent years, various studies have been conducted to address the issue of reduced communication due to mask use, which has become unavoidable due to COVID-19. Some studies used a wearable display to produce lip motion and facial expressions. In this research, sensors were inserted in the mask to recognize the contour of the lips. The display device allows other people to view the shape of mouth or emotion of the person [Lee et al. 2020; Nam et al. 2020]. The appearance of the mouth generated in this research differs from the genuine mouth shape because the mouth shape was created using basic graphics. Another study examines the shape of a speaker's mouth using voice. This study developed a system that projects lip animations onto mask regions. This study was conducted through the analysis of sound and the creation of lip-sync animations [Takagi et al. 2021]. However, it is suitable for use in a limited situation (e.g., a dark environment) because this is a projection method.

3 IMPLEMENTATION

The overall system design is shown in Figure 2. This technology is mainly divided into two modules. The first module uses a visual source. This part recognizes the user's facial pose, the mask's shape, and emotion. The second module uses an audio source. This module analyzes user's speech to generate a lip-sync animation. The final video is created by combining the information from the previous processes. This project was implemented in Unity 2019.3.15¹.

3.1 Face Detection and Recognizing a Mask

To synthesize the facial animation at the proper position, we detected head pose and facial landmarks using the Face Analyzer tool². Face Analyzer can extract the head pose and 68 facial landmarks based on deep learning. Using facial landmarks, this system can recognize the lower part of the face.

3.2 Emotion recognition and Expression

This project provides a function for recognizing emotions in a face based on information from the upper part of the face and synthesizing a facial expression corresponding to the recognized emotion. We used the Face analyzer tool to detect user's emotions. There are six categories of expressions that are commonly used to express emotions [Ekman 1992]. We chose four categories and added a neutral category. As a result, emotion categories contain Happy, Angry, Sad, Surprise, and Neutral. Also, we created five

facial animations to express user's emotion (see Figure 1). Emotion animation is fused with Lip-sync animation, described in the next subsection.

3.3 Lip-sync Animation

Lip-sync animation is created using Oculus Lip-sync library. This module recognized fifteen phonemes and represented lip-sync animation using blend shapes that correlate to viseme analysis in real-time³.

3.4 Synthesizing Animation

Finally, the system renders lower facial animation in an appropriate position on the face. To render the correct position, we render the avatar in a position corresponding to the mask shape extracted from section 3.2. Emotional animation and lip-sync animation are controlled in the form of blendshapes. The final rendering results are shown in Figure 2.

4 CONCLUSION AND FUTURE WORK

This system aims to improve communication by artificially synthesizing lower-face animation and augmenting the face mask. This project includes developing a system that generates lip-sync animation based on the user's voice and emotional facial animation based on the user's upper face information. In addition, we proposed techniques for lower-part face animation synthesis on the real face.

We are interested in research to improve rendering quality so that the lower face animation will combine naturally with the real face. Furthermore, emotion recognition accuracy is insufficient because only upper-face information is used to recognize the user's emotions. Therefore, we are interested in boosting the accuracy of using deep learning. And also, we will expand the number of available facial expressions to support a broader range of emotions. A qualitative and quantitative user study will be conducted in the future to determine whether the technology utilized in the study genuinely increases communication efficiency.

ACKNOWLEDGMENTS

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¹<https://unity.com/>

²<https://www.illuni.com>

³<https://developer.oculus.com/documentation/unity/audio-ovrlipsync-unity/>